ZEROCRUMB: A WEB PLATFORM FOR RESCUING SURPLUS FOOD AND CONNECTING DONORS, NGOs AND COMMUNITIES

PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this Project Report titled "ZEROCRUMB: A WEB PLATFORM FOR RESCUING SURPLUS FOOD AND CONNECTING DONORS, NGOs AND COMMUNITIES" is the bonafide work of MADHANESH M (241701026) and NIRANJAN S (241701040), students of the Department of Computer Science and Design, Rajalakshmi Engineering College, submitted in partial fulfillment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Design.

The project explores a web platform that connects food providers, NGOs and recipient organizations to rescue surplus food and improve access to meals. Its intended social contribution relates to SDG 2 (Zero Hunger), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), and SDG 17 (Partnerships for the Goals), with primary emphasis on SDG 2 and SDG 12 through reducing avoidable food waste and improving access to food.

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ABSTRACT

ZEROCRUMB is a proposed web platform that connects food providers, NGOs and recipient organizations to rescue surplus food and reduce avoidable food waste. The project addresses a coordination gap: edible surplus from hostels, restaurants, caterers and events may be discarded because donors lack a quick, trusted way to find recipients, while orphanages and community organizations struggle to plan reliable, nutritious meals. Following a design-thinking approach, defines the problem, studies stakeholders and their pain points, frames the design challenge of minimizing food waste while helping orphanages receive nutritious surplus food efficiently, and proposes a four-step workflow of listing, matching, claiming and delivering. Planned features include donation listings, location matching, real-time alerts, verified coordination, a safety checklist and an impact dashboard, supported by a shared database, location services, notifications and cloud hosting. Prototype screens for an NGO coordinator dashboard and a food provider dashboard illustrate the interface concept. The expected outcomes are more surplus food identified before it spoils, faster matching between donors and recipients, and better visibility of donation and delivery status; these are expected benefits, not measured results. ZEROCRUMB supports coordination and does not replace food-safety checks or responsible human handling.

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TABLE OF CONTENTS

CHAPTER / SECTION	TITLE	PAGE
	BONAFIDE CERTIFICATE	i
	ABSTRACT	ii
	ACKNOWLEDGEMENT	iii
	LIST OF TABLES	iv
	LIST OF FIGURES	v
	LIST OF ABBREVIATIONS	vi
CHAPTER 1	**INTRODUCTION**	1
1.1	General	1
1.2	Background	1
1.3	Problem Statement	2
1.4	Motivation	2
1.5	Objectives	2
1.6	Scope	3
1.7	Existing System and Limitations	3
1.8	Proposed System and Advantages	4
CHAPTER 2	**LITERATURE REVIEW**	5

CHAPTER / SECTION	TITLE	PAGE
2.1	Introduction	5
2.2	Review of Related Work	5
2.3	Comparative Analysis	6
2.4	Research Gap	7
CHAPTER 3	**THEORETICAL BACKGROUND AND SYSTEM DESIGN**	8
3.1	Core Technologies	8
3.2	System Architecture and Data Flow	9
3.3	Use Case, Activity and Conceptual Classes	10
CHAPTER 4	**METHODOLOGY AND PROJECT DESCRIPTION**	11
4.1	Requirements	11
4.2	Technology Selection and Modules	12
4.3	API Endpoints and Feasibility	13
4.4	Testing and Initial Results	14
CHAPTER 5	**CONCLUSION AND FUTURE ENHANCEMENTS**	15
	REFERENCES	17
	CO–SDG RELEVANCE RECORD	18
	COURSE OUTCOMES	19

CHAPTER / SECTION	TITLE	PAGE
	PROGRAM OUTCOMES	20
	PROGRAM SPECIFIC OUTCOMES	21
	CO–PO–PSO MAPPING	22

LIST OF TABLES

TABLE NO.	TITLE
2.1	Comparative Analysis of Related Approaches
3.1	Conceptual Classes
4.1	Functional Requirements
4.2	Non-Functional Requirements
4.3	Technology Selection
4.4	API Endpoints
4.5	Testing Methodology
5.1	Phase II Schedule
A.1	CO–SDG Relevance Record
A.2	CO–PO–PSO Mapping

LIST OF FIGURES

FIGURE NO.	TITLE
3.1	ZEROCRUMB High-Level System Architecture
3.2	Donation Data Flow
3.3	Use Case Overview
3.4	Activity Flow for Food Donation

LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION
API	Application Programming Interface
DFD	Data Flow Diagram
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
NGO	Non-Governmental Organization
SDG	Sustainable Development Goal
UI	User Interface
UX	User Experience

CHAPTER 1: INTRODUCTION

1.1 General

Food waste and food insecurity can exist side by side in the same community. Hostels, restaurants, caterers and events may produce edible surplus that is discarded because donors do not have a quick, trusted way to find recipients, while orphanages and community organizations can struggle to plan reliable, nutritious meals. ZEROCRUMB investigates a web platform that connects food providers, NGOs and recipient organizations so that safe surplus food can be listed, discovered, claimed and delivered.

1.2 Background

Food donation is time-sensitive: surplus food must be collected promptly, with freshness, hygiene and transit conditions considered. In practice the process often depends on informal calls and personal contacts, which makes availability, collection and delivery status difficult to track. A shared system can make listings visible to nearby NGOs and coordinators. A design-thinking approach of empathising, defining, ideating and prototyping helps ground the solution in the needs of donors, coordinators and recipients.

1.3 Problem Statement

Large quantities of edible surplus from hostels, restaurants, caterers and events may be thrown away while nearby orphanages and communities need food. Without a shared system, donors and recipients rely on informal calls and personal contacts, so donors may not know who needs food or who can collect it. The project problem is to design a platform that makes the donation journey visible, simple and time-aware while keeping food-safety checks explicit.

1.4 Motivation

A small improvement in coordination can create value for donors, NGOs and recipients: more meals reach people in need and less food is wasted. The guiding question is how technology can make donation easier to act on and to trust. The project also provides an

applied study of design thinking, stakeholder research, web application design, and responsible social-impact technology.

1.5 Objectives

The objectives are to reduce the wastage of edible surplus food; connect donors with orphanages and needy communities; improve pickup and delivery coordination; encourage responsible food donation practices; and track impact through meals and deliveries. Expected outcomes include more surplus food identified before it spoils, faster matching between donors and recipients, and better visibility of donation and delivery status.

1.6 Scope

Included scope: a Phase I design-thinking study, stakeholder and pain-point analysis, a proposed web platform with donation listings, location matching, alerts, a safety checklist and an impact dashboard, and prototype screens for the NGO coordinator and food provider dashboards. Limitations: the platform coordinates donations but does not replace food-safety checks, eligibility verification or responsible human handling. Adoption, volunteer and transport availability, location accuracy, and local regulatory requirements may affect real-world use. A working deployment and field validation are outside the current phase.

1.7 Existing System and Limitations

Food is commonly redistributed through phone calls, personal contacts, messaging groups and NGO-run collection drives. These methods can work for known contacts but offer little shared visibility of who needs food, who can collect it, and the status of a pickup. Donors may be uncertain about collection time and food freshness, while NGOs and orphanages face unpredictable availability, scattered coordination and difficult transport timing. Limited tracking also makes the final impact hard to see.

1.8 Proposed System and Advantages

ZEROCRUMB proposes a coordinated rescue network: providers list surplus food, nearby NGOs discover it, an eligible organization claims it, and pickup and delivery are tracked. Shared listings, location matching,

real-time alerts, a safety checklist and an impact dashboard reduce manual steps and keep the process transparent, while human checks remain responsible for food safety.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

Work relevant to ZEROCRUMB spans design-thinking methodology, food-waste and food-security studies, food-redistribution practices run by NGOs and volunteer networks, and digital coordination platforms. The task matters: listing surplus food, matching it with a nearby recipient and delivering it safely are distinct problems that a platform must connect.

2.2 Review of Related Work

Design thinking is a human-centred, iterative approach commonly described through the stages of empathise, define, ideate, prototype and test. It encourages teams to understand stakeholders before proposing technology, which suits a social-welfare problem such as surplus food redistribution.

Global assessments such as the UNEP Food Waste Index Report and FAO's food security reports document large-scale food waste alongside persistent food insecurity. Local figures must be verified before being reported as evidence.

Food is commonly redistributed through NGO networks, volunteer groups, food banks and surplus-food applications. Effectiveness depends on timely information and reliable pickup and delivery. Features and coverage of specific platforms must be verified before comparative claims are made.

Food-safety guidance, such as that issued by the Food Safety and Standards Authority of India (FSSAI), emphasises hygienic handling and prompt use of food. A donation platform should therefore treat freshness and hygiene checks as part of the workflow; the exact regulatory requirements must be validated against current official guidance.

APPROACH	STRENGTH	LIMITATION	RELEVANCE TO ZEROCRUMB
Informal calls and personal contacts	Quick for known donors and recipients	No shared visibility or tracking	Problem baseline
Messaging groups and social media	Wide reach with little setup	Unstructured; claims and status hard to track	Complementary channel
NGO-run collection drives	Established volunteer networks	Limited by manual coordination and capacity	Key stakeholder workflow
Surplus-food sharing applications	Structured listings and matching	Focus and coverage vary by region and recipient	Reference model for the platform
Coordinated web platform (ZEROCRUMB)	Listings, matching, alerts and tracking in one place	Needs adoption and human safety checks	Phase I proposed direction

2.4 Research Gap

A useful gap for this project is the combination of listing, location-aware matching, safety prompts, delivery tracking and impact reporting in one coordinated workflow aimed at orphanages and community organizations. The gap is not that food donation platforms are wholly absent; rather, the project investigates how a donor-to-recipient workflow can be made simple, time-aware and transparent for local stakeholders. Large-scale adoption, verified food-safety compliance, real-world pilots and user studies remain open work.

CHAPTER 3: THEORETICAL BACKGROUND AND SYSTEM DESIGN

3.1 Responsive Web Interface

The platform is planned as a responsive web interface that works on phones and desktops, with separate views for food providers and NGO coordinators. The NGO coordinator dashboard presents the donation feed, active runs and emergency requests, while the food provider dashboard presents surplus listings, donation history and impact insights. Forms should capture food type, quantity and best-before time with minimal effort, and screens should make status and time remaining clear.

3.2 Shared Database

A shared database stores users, organizations, donation listings, claims, pickup status and delivery records. A consistent data model links each listing to its provider, claiming organization and delivery status so that all users see the same current information. Listing times and best-before times should be stored to support time-aware matching, and personal or location data should be limited to what the workflow requires.

3.3 Location Services and Matching

Location services support nearby matching and routing. Listings carry a pickup location and a best-before time, so coordinators can see opportunities that are close enough to collect in time. Matching should consider distance, quantity and time remaining; the displayed order is a convenience for coordinators, and the claiming organization remains responsible for deciding whether it can safely collect and use the food.

3.4 Backend Services and Notifications

The backend service layer sits between the web interface and the shared database. It validates requests, applies role-based rules, manages claims, updates pickup and delivery status, and triggers notifications. The service should include input validation,

error handling, health checks, access control and appropriate logging.

3.5 Cloud Hosting

Cloud hosting makes the web application accessible to all stakeholders. Deployment should use secure transport, documented configuration, data backups and monitoring. Hosting choices, costs and any region-specific requirements must be confirmed before deployment. The interface should show food-safety prompts and avoid implying that listed food has been inspected or certified by the platform.

3.6 System Architecture and Data Flow

The proposed architecture consists of a provider interface, a web platform for listing and matching, a shared database for users and donations, an NGO and volunteer interface for claiming and delivering, and supporting location, notification and cloud-hosting services. Data flows from a listing to matching and alerts, then to a claim, pickup and delivery handoff. A production version should minimise personal data and use secure transport.

3.7 Data Flow Diagrams Overview

Context-level DFD: food providers, NGOs and volunteers, and recipients interact with the ZEROCRUMB system, which uses database, location and notification services. Level-1 DFD: listing creation, validation and storage, location matching, alert generation, claiming, pickup and delivery tracking, and impact reporting are separate processes. These diagrams should be drawn using the final implementation details before submission.

3.8 Use Case and Activity Descriptions

Primary use case: a food provider creates a listing with food type, quantity and best-before time; a nearby NGO coordinator receives an alert and claims it. Activity flow: create listing; validate and store it; notify

nearby coordinators; claim; complete the safety checklist and collect; confirm handoff; update the dashboard; handle expired listings.

3.9 Conceptual Classes

CLASS / COMPONENT	RESPONSIBILITY	KEY DATA
FoodProvider	Creates and manages surplus listings	Provider profile, listing history
DonationListing	Describes available surplus food	Food type, quantity, best-before time
MatchingService	Finds nearby suitable listings	Location, time window, status
ClaimManager	Assigns a listing to an eligible organization	Organization, claim time, listing status
DeliveryTracker	Records pickup and handoff progress	Pickup status, handoff time
NotificationService	Informs users of new listings and updates	Recipient, message, timestamp

CHAPTER 4: METHODOLOGY AND PROJECT DESCRIPTION

4.1 Requirement Analysis

Requirements were derived from the stakeholder pain points identified during the empathise and define stages: providers must be able to list surplus food quickly, NGOs must discover and claim nearby listings, and all parties need visible pickup and delivery status. Quality requirements include responsiveness, maintainability, privacy, fault handling, and food-safety awareness.

4.1.1 Functional Requirements

ID	REQUIREMENT
FR1	The system shall let providers create surplus food listings.
FR2	The system shall allow NGOs and coordinators to view nearby available listings.
FR3	The system shall allow an eligible organization to claim a listing.
FR4	The system shall send alerts about new listings and status updates.
FR5	The system shall validate requests and return structured responses.
FR6	The system shall track pickup and delivery status.
FR7	The interface shall provide NGO and provider dashboards.
FR8	The system shall show an impact summary of meals and deliveries.

4.1.2 Non-Functional Requirements

ID	REQUIREMENT
NFR1	Responsiveness: listings, alerts and status updates should minimise noticeable delay.
NFR2	Reliability: invalid inputs and temporary failures should not lose listings or claims.
NFR3	Maintainability: listing, matching, notification, tracking, API, and UI should remain modular.
NFR4	Privacy: personal and location data should be limited to what is needed and protected.
NFR5	Usability: listing, claiming, and status messages should be simple and understandable.
NFR6	Portability: dependencies and configuration should be documented.
NFR7	Security: API traffic and access controls should be configured appropriately.

4.2 Technology Selection

TECHNOLOGY	ROLE	REASON FOR SELECTION
Responsive Web Interface	Provider and NGO dashboards	Accessible on phones and desktops without installation
Shared Database	Users, listings and deliveries	Single source of truth for listings and status
Location Services	Nearby matching and routing	Connects listings with nearby recipients
Notification Service	Real-time alerts	Keeps users informed about new listings and updates

Backend API	Listing and claim workflow	Validated request handling and

TECHNOLOGY	ROLE	REASON FOR SELECTION
		API documentation
Cloud Hosting	Deployment	Makes the web application accessible to users

4.3 System Modules

4.3.1 Provider and Listing Module

Allows food providers to create and manage surplus listings with food type, quantity and best-before time, and to view donation history and impact insights. The module validates required fields and prevents expired listings from being offered.

4.3.2 Matching and Claim Module

Uses location and time information to show nearby listings to NGOs and coordinators, sends alerts, and lets an eligible organization claim a listing. Claim rules should prevent duplicate claims.

4.3.3 Tracking and Dashboard Module

Tracks pickup progress and delivery handoff, presents the safety checklist, and summarises meals and deliveries on the dashboards. Network errors, expired listings and unavailable services should result in clear user-facing messages.

4.4 API Endpoints

METHOD / PATH	PURPOSE	EXPECTED RESPONSE
GET /health	Check service availability	Service status and version
POST /donations	Create a surplus food listing	Listing ID and status
GET /donations/nearby	List available listings near an organization	Listings with quantity and best-before time

POST /donations/{id}/claim	Claim a listing for collection	Claim status

These endpoints describe the planned API contract. They should be aligned with the implemented application before claiming that every endpoint is available.

4.5 Feasibility Study

Economical Feasibility

The prototype can be developed primarily with open-source software and existing computers and mobile phones. Costs may arise from cloud hosting, location and notification services, testing devices, or deployment. A final cost estimate should be prepared from the actual hosting and service choices.

Technical Feasibility

Responsive web frameworks, a shared database, location services and notification services provide compatible building blocks for a prototype. Technical risks include location accuracy, timely alerts, data consistency, handling simultaneous claims, and cloud deployment constraints. These risks require staged testing.

Social Feasibility

The project targets reduced food waste and improved access to food. It should be developed with input from donors, NGOs, volunteers and orphanage administrators, respect local food-handling practices, and avoid presenting the platform as a replacement for food-safety checks and responsible human judgement.

4.6 Testing Methodology

TEST LEVEL	PURPOSE	EXAMPLE CHECK
Unit testing	Verify isolated components	Listing field validation and expiry handling
Integration testing	Verify connected components	Client-to-API request and response
Functional testing	Verify required user actions	Create listing, claim, update status, recover from errors

TEST LEVEL	PURPOSE	EXAMPLE CHECK
System testing	Evaluate complete workflow	Provider-to-recipient workflow behaviour and alert timing

4.7 Initial Results

At the Phase I stage, the design-thinking artefacts and prototype screen concepts have been produced: a problem statement, stakeholder and pain-point analysis, the design challenge, the proposed list–match–claim–deliver workflow, and dashboard concepts for NGO coordinators and food providers. No quantitative performance or impact results have been recorded yet. The expected outcomes are intended benefits and should be evaluated in Phase II using test cases, user feedback and documented prototype versions.

4.8 Challenges

Key challenges include adoption by donors and NGOs, verifying the eligibility and trustworthiness of participants, keeping listings accurate and current, short best-before windows, transport and volunteer availability, location accuracy, and food safety and hygiene during transit. Phase II should measure each issue through user feedback and realistic testing rather than relying only on assumptions.

CHAPTER 5: CONCLUSION AND FUTURE ENHANCEMENTS

5.1 Conclusion

ZEROCRUMB investigates a coordinated web platform that connects food providers, NGOs and recipient organizations to reduce avoidable food waste and improve access to surplus food. Phase I defines the problem, studies stakeholders and their pain points, sets design principles, proposes the solution workflow and features, specifies requirements and architecture, and presents prototype dashboard concepts. The current evidence supports a design and prototype direction only; it does not establish real-world impact or production readiness.

5.2 Current Phase-I Status

The Phase I deliverables include the problem statement, stakeholder and pain-point analysis, the design challenge, the proposed solution and features, a conceptual architecture, prototype dashboard concepts, requirements, and feasibility considerations. Before formal evaluation, the team should preserve the design artefacts, prototype versions, test cases, and screenshots supporting reported results.

5.3 Limitations

The prototype is limited to the conceptual design and screen concepts of Phase I. Technology coordinates donations, but food safety, eligibility and handling still require human checks. Impact statements are expected outcomes, not measured results, and real-world performance requires end-to-end verification.

5.4 Phase-II Objectives

Phase II will focus on validating requirements with stakeholders, building the working web platform with listing, matching, claiming and delivery tracking, integrating location and notification services, usability testing, privacy and security review, and cloud deployment. Acceptance thresholds should be defined before final evaluation.

PHASE-II PERIOD	ACTIVITY	DELIVERABLE
Weeks 1–2	Validate requirements and finalise the data model	Refined requirements and database design
Weeks 3–4	Build provider and NGO listing and claim flows	Working listing and claim modules
Weeks 5–6	Integrate location matching and real-time alerts	Nearby matching and notifications working
Weeks 7–8	Implement tracking, safety checklist, dashboard	End-to-end prototype
Weeks 9–10	Measure performance and conduct user-centred review	Test results and feedback summary
Weeks 11–12	Fix defects and prepare final documentation/demo	Final Phase II submission

5.5 Future Enhancements

ENHANCEMENT	EXPECTED BENEFIT
Expanded donor and recipient network	Broader coverage of surplus food rescue
Route planning for pickups	Faster collection of time-sensitive food
Verified organization onboarding	Greater trust between donors and recipients
Food-safety guidance and checks	Safer handling of donated food
Demand and surplus forecasting	Better planning of meals for recipients
Privacy-preserving data handling	Reduced exposure of personal and location data
Mobile app with push alerts	Faster response to new listings
Multilingual interface	Easier use across local languages

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CO–SDG RELEVANCE RECORD

SDG	TITLE	RELEVANCE TO ZEROCRUMB	PRIORITY
SDG 2	Zero Hunger	Connects surplus food with orphanages and communities facing food insecurity.	Primary
SDG 11	Sustainable Cities and Communities	Strengthens local coordination among donors, NGOs and communities.	Supporting
SDG 12	Responsible Consumption and Production	Reduces avoidable food waste by redirecting edible surplus in time.	Primary
SDG 17	Partnerships for the Goals	Brings food providers, NGOs and volunteers together on one platform.	Supporting

COURSE OUTCOMES (CO1–CO5)

CO	OUTCOME STATEMENT
CO1	Apply foundational computing, mathematics, and engineering knowledge to define a technical problem.
CO2	Analyse the food-redistribution problem and evaluate suitable stakeholder needs and solution approaches.
CO3	Design a modular donation-coordination solution considering functional requirements and constraints.
CO4	Conduct tests, interpret prototype feedback, and evaluate limitations using appropriate evidence.
CO5	Use modern software and digital tools responsibly while considering food safety, privacy, and societal impact.

PROGRAM OUTCOMES (PO1–PO12)

PO	OUTCOME
PO1	Engineering knowledge
PO2	Problem analysis
PO3	Design/development of solutions
PO4	Conduct investigations of complex problems
PO5	Modern tool usage
PO6	The engineer and society
PO7	Environment and sustainability
PO8	Ethics
PO9	Individual and team work
PO10	Communication
PO11	Project management and finance
PO12	Life-long learning

PROGRAM SPECIFIC OUTCOMES (PSO1–PSO4)

PSO	PROGRAM-SPECIFIC OUTCOME
PSO1	Apply computing and design knowledge to develop user-centred software solutions.
PSO2	Use data structures, algorithms, databases, and software engineering practices to solve application problems.
PSO3	Integrate modern AI and software tools to build and evaluate interactive systems.
PSO4	Design inclusive digital experiences while considering usability, privacy, and societal needs.

CO–PO–PSO MAPPING

Mapping scale: 1 – Low, 2 – Medium, 3 – High. Ratings below are a proposed project-level mapping and should be reviewed against the department's officially approved CO/PO/PSO definitions before submission.

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PS O1	PS O2	PS O3	PSO 4
CO 1	3	2	1	1	1	1	0	0	1	1	0	2	2	1	1	1
CO 2	2	3	2	3	2	1	0	1	0	1	0	2	2	2	2	1
CO 3	2	2	3	2	3	2	1	2	1	2	2	1	2	3	3	2
CO 4	2	3	2	3	2	1	0	1	0	1	0	2	2	2	2	1
CO 5	1	1	2	1	3	3	1	3	2	2	1	2	1	2	2	3